

Theta Closure in Modal Triplet Theory IV: Conditional Gravity Scaling and Cosmological Cutoff Audit

Peter Nero

July 2026

Abstract

We re-evaluate the propagation of the gauge-profile geometry into gravity and cosmology. Within the auxiliary six-dimensional $S^1 \times S^2 \times \text{Nil}_3$ product ansatz of Papers I–III, the updated dimensionless internal-volume coefficient is $\hat{V}_{\text{int}} = 20.07064 R_1^3$. Restoring the common internal length ℓ_{int} gives $G_N^{-1} = 20.07064 \ell_{\text{int}}^6 R_1^3 / G_{10}$ if that auxiliary manifold is adopted as the compactification. It is not the selected q79/Fu–Yau space, so the coefficient is an ansatz diagnostic rather than the current MTT internal volume. Even within the ansatz, the gauge profile does not determine Newton’s constant without an absolute scale and fundamental gravity normalization. A gauge matching point does not by itself select the physical cutoff Λ_Θ . The correctly normalized conditional relation is $r \leq 2\epsilon^2(\Lambda_\Theta/M_{\text{Pl}})^2/(\pi^2 A_s)$ when $H \leq \epsilon\Lambda_\Theta$. The paper therefore supplies conditional scaling laws and an assumption audit, not cross-sector numerical closure.

Revision note for this edition

Supersedes. *Theta Closure in Modal Triplet Theory IV: Gravity and Cosmology from the Closure Scale*, first edition.

Reason. A dimensionless auxiliary volume and the obsolete few-TeV gauge crossing were promoted into Newton and primordial-tensor estimates without an absolute length, G_{10} , or selected physical compactification.

Resolution. Version 2 restores dimensions, confines the coefficient $20.07064 R_1^3$ to the auxiliary $S^1 \times S^2 \times \text{Nil}_3$ ansatz, and withdraws the numerical tensor claim.

Retained result. The dimensional-reduction and cutoff inequalities survive as conditional scaling relations.

Remaining boundary. A q79/Fu–Yau volume, absolute scale, gravity normalization, and cosmological solution must be selected independently.

1 Introduction

Papers I–III now establish a selected common-scheme gauge profile, a calibrated leading geometric realization, and a conditional twistor representation check. They do not select an absolute internal length, a fundamental ten-dimensional gravitational coupling, or a cosmological coherence cutoff. This distinction controls what can be transported into the gravity and cosmology sectors.

This paper asks two scoped questions. First, what dimensionless auxiliary volume coefficient follows if the effective round- $S^2/\text{nilmanifold}$ product ansatz is used? Second, what tensor inequality follows if a separately selected coherence cutoff bounds the Hubble scale? The answers are conditional scaling relations. They must not be promoted into predictions of G_N or r until the missing absolute-scale and cutoff source theorems are supplied.

2 Central picture and dependency flow

The central picture is dimensional analysis with two independent exits. Gauge ratios can constrain dimensionless shape data, but Newton’s constant also depends on an absolute internal length and the higher-dimensional gravitational normalization. Likewise, a renormalization scale used to quote gauge couplings is not a physical ultraviolet or coherence cutoff.

The two calculations therefore have the form

$$\begin{aligned} \text{calibrated auxiliary shape} &\longrightarrow \widehat{V}_{\text{aux}} + \{\ell_{\text{int}}, G_{10}\} \longrightarrow G_N, \\ \{\Lambda_\Theta, \epsilon, \text{cosmological state}\} &\longrightarrow r \text{ bound.} \end{aligned}$$

The first line is evaluated in Section 3 and the second in Section 4. In both lines the braces contain inputs that the gauge-profile calculation does not select.

3 Auxiliary volume and conditional Newton scaling

We compute the gravity scaling implied by the auxiliary effective product ansatz. The calculation introduces no retuning of the dimensionless profile, but it retains the independent absolute length ℓ_{int} and fundamental coupling G_{10} .

3.1 Gravitational coupling in Modal Triplet Theory

In Modal Triplet Theory, the effective four-dimensional gravitational coupling arises from dimensional reduction of the higher-dimensional coherent sector. At leading order this takes the form

$$\frac{1}{G_N} = \frac{\text{Vol}(X_{\text{int}})}{G_{10}}, \quad (3.1)$$

where:

- G_{10} is the fundamental gravitational coupling of the modal theory,
- X_{int} is the internal coherent space.

Equation (3.1) is the standard unwarped dimensional-reduction form assumed here [1]. Its applicability to MTT requires the product metric, Einstein-frame convention, and absence or control of warp/dilaton corrections.

3.2 Internal volume in terms of Θ

Define the auxiliary six-dimensional product

$$X_{\text{aux}} = S_{\text{cen}}^1 \times \Sigma_2 \times \Sigma_3, \quad (3.2)$$

where:

- S_{cen}^1 is the central circle of radius R_1 ,
- Σ_2 is the lens layer,
- Σ_3 is the color layer $\Gamma \backslash \text{Nil}_3$.

Here Σ_2 is the two-dimensional effective lens base, not the three-dimensional lens space. Thus the dimension is $1 + 2 + 3 = 6$. This is a declared auxiliary coordinate model. In the general MTT carrier the shared circle is common $U(1)$ phase/holonomy data and need not be an independent product coordinate; the current q79/Fu–Yau compactification is a different global manifold.

After factoring out a common physical length ℓ_{int} , write the remaining radii and metric parameters as dimensionless quantities. Under the unwarped product ansatz the auxiliary volume factorizes as

$$\text{Vol}(X_{\text{aux}}) = \ell_{\text{int}}^6 (2\pi R_1) \widehat{\text{Area}}(\Sigma_2) \widehat{\text{Vol}}(\Sigma_3). \quad (3.3)$$

3.3 Insertion of the calibrated gauge-profile geometry

Paper II's calibrated leading ansatz gives the dimensionless relations

$$\widehat{\text{Area}}(\Sigma_2) = 4\pi(f_2 R_{\text{lens}})^2 = 4\pi(0.2555137 R_1), \quad (3.4)$$

$$\widehat{\text{Vol}}(\Sigma_3) = c = 0.9948493 R_1, \quad (a = b = 1). \quad (3.5)$$

These are profile-realization data, not an independent gravity-sector selection. Substitution gives

$$\widehat{V}_{\text{int}} = (2\pi R_1) [4\pi(0.2555137 R_1)] (0.9948493 R_1) \quad (3.6)$$

$$= 20.0706400 R_1^3, \quad (3.7)$$

and therefore

$$\boxed{\text{Vol}(X_{\text{aux}}) = 20.0706400 \ell_{\text{int}}^6 R_1^3.} \quad (3.8)$$

3.4 Conditional expression for G_N

If, in addition to the dimensional-reduction assumptions, one identifies the physical internal space with X_{aux} , then

$$\boxed{\frac{1}{G_N} = \frac{20.0706400 \ell_{\text{int}}^6 R_1^3}{G_{10}}.} \quad (3.9)$$

The gauge profile fixes only the displayed dimensionless shape coefficient within the chosen ansatz. It does not fix $\ell_{\text{int}}^6/G_{10}$, this paper does not derive the reduction formula from a selected MTT action, and the required identification $X_{\text{int}} = X_{\text{aux}}$ is not established. Thus no numerical prediction of Newton's constant follows.

3.5 Worked example and interpretation: one shape, an unresolved scale family

Set $R_1 = 1$ inside the auxiliary model. The gauge-profile calibration then fixes only

$$\widehat{V}_{\text{aux}} = 20.0706400, \quad \frac{1}{G_N} = 20.0706400 \frac{\ell_{\text{int}}^6}{G_{10}}.$$

For any $s > 0$, the simultaneous change

$$\ell_{\text{int}} \mapsto s \ell_{\text{int}}, \quad G_{10} \mapsto s^6 G_{10}$$

leaves G_N unchanged. Conversely, holding G_{10} fixed while changing ℓ_{int} changes the predicted Newton constant by the sixth power. This explicit degeneracy is why a dimensionless overlap fit cannot determine four-dimensional gravity without an absolute-scale theorem.

4 Conditional cosmological cutoff relation

The old gauge crossing did not select a physical coherence cutoff. In particular, the former assignment $\Lambda_\Theta \sim 5$ TeV and the scan $[3, 10]$ TeV are withdrawn. The current gauge matching point $Q = M_t$ is a renormalization convention and must not be identified with Λ_Θ .

Let a future source theorem select a physical cutoff Λ_Θ , and suppose the relevant cosmological solution satisfies the quantitative admissibility condition

$$H \leq \epsilon \Lambda_\Theta, \quad 0 < \epsilon < 1. \quad (4.1)$$

For vacuum tensor fluctuations in standard slow-roll normalization [2],

$$P_t = \frac{2H^2}{\pi^2 M_{\text{Pl}}^2}, \quad r = \frac{P_t}{A_s}. \quad (4.2)$$

Here $M_{\text{Pl}} = (8\pi G_N)^{-1/2}$ is the reduced Planck mass and A_s is the scalar curvature-perturbation amplitude in the same convention. Consequently,

$$r \leq \frac{2\epsilon^2}{\pi^2 A_s} \left(\frac{\Lambda_\Theta}{M_{\text{Pl}}} \right)^2. \quad (4.3)$$

This corrects the legacy formula, which omitted the factor $2/(\pi^2 A_s)$. Equation (4.3) remains conditional on the tensor-production model as well as on independently selected values of Λ_Θ and ϵ . It supplies no current numerical bound.

5 Status and falsifiability

The conditional relation (4.3) can become falsifiable only after MTT independently selects a cutoff, an admissibility margin, and a cosmological production law. A measured tensor amplitude could then test that joint hypothesis. Without those selections, neither detection nor non-detection of primordial tensors presently falsifies the gauge-profile closure program. The withdrawn 10^{-30} – 10^{-29} interval must not be quoted as an MTT prediction.

6 Conclusion

Within the auxiliary product ansatz, the updated gauge profile fixes the dimensionless ansatz-volume coefficient to $20.0706400 R_1^3$. Restoring dimensions shows explicitly why this is not yet a prediction of Newton’s constant: the independent combination $\ell_{\text{int}}^6/G_{10}$ remains, and no theorem identifies X_{aux} with the selected q79/Fu–Yau compactification.

The cosmological result is likewise a conditional scaling law. The obsolete few-TeV gauge crossing cannot serve as a physical coherence cutoff, and the legacy numerical tensor bound is withdrawn. A future closure theorem must select Λ_Θ , the quantitative margin ϵ , and the applicable cosmological state before Equation (4.3) becomes a numerical prediction. Paper IV therefore documents cross-sector dependencies and correct formulas; it does not establish gravity or cosmology closure.

Computational Evidence and Reproducibility

This paper audits possible gravity and cosmology scaling from gauge-profile geometry. None of the mapped Standard Model packets derives a physical gravitational normalization or cosmological

cutoff, so all closed rows below are corpus-state cross-checks rather than direct proof. The strict upgrade remains open.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper's local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Corpus-state cross-checks.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A01/direct_k_higgs_row (*derived exact*).
Promoted direct K_threshold.Omega_H.lambda row.
- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.
- A01/strict_pew_row (*derived exact*).
Promoted P_EW source row at the declared one-shared-primitive standard.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

- [1] M. J. Duff, B. E. W. Nilsson, and C. N. Pope, *Kaluza–Klein supergravity*, Physics Reports **130** (1986) 1–142. [https://doi.org/10.1016/0370-1573\(86\)90163-8](https://doi.org/10.1016/0370-1573(86)90163-8)
- [2] D. Baumann, *TASI lectures on inflation*, <https://arxiv.org/abs/0907.5424>
- [3] P. Nero, *Modal Triplet Theory: Admissibility, Encodings, and the Structure of Physical Description*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255621>
- [4] P. Nero, *Modal Triplet Theory: Foundation*, Zenodo preprint, September 2025. <https://doi.org/10.5281/zenodo.16949762>
- [5] P. Nero, *Fixed Points I–VI: Complete Coherence Spine*, Zenodo preprints, August 2025. <https://doi.org/10.5281/zenodo.16948748>
- [6] P. Nero, *The Projection–Admissibility Principle: Structural Constraints on Effective Physical Description*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255838>
- [7] P. Nero, *Closure and Inevitability in Modal Triplet Theory*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255510>
- [8] P. Nero, *Coherence Capacity as the Fundamental Resource of Effective Physics*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255905>
- [9] P. Nero, *Dynamics of Coherence Capacity: Transport, Concentration, and Exhaustion*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18256048>
- [10] P. Nero, *Modal Triplet Theory: From MTT to Quantum Mechanics*, Zenodo preprint, September 2025. <https://doi.org/10.5281/zenodo.17074246>
- [11] P. Nero, *From MTT to Quantum Field Theory*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17068816>
- [12] P. Nero, *Modal Triplet Theory: From MTT to General Relativity*, Zenodo preprint, October 2025. <https://doi.org/10.5281/zenodo.16950597>
- [13] P. Nero, *Modal Triplet Theory: From MTT to a UV-Finite, Unitary Quantum Gravity*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17077671>
- [14] P. Nero, *Measurement as Disturbance and Stabilization in Modal Triplet Theory*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17177404>
- [15] P. Nero, *Projection, Probability, and Irreversibility: Shadow Bridges Between Measurement, Black Holes, and Cosmology in Modal Triplet Theory*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18256408>

- [16] P. Nero, *Modal Fixed Points, Bell's Beables, and the Limits of Factorization*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17076300>
- [17] P. Nero, *Temporal Bell Inequalities and Global Consistency in Modal Triplet Theory*, Zenodo preprint, August 2025. <https://doi.org/10.5281/zenodo.18208884>
- [18] P. Nero, *Modal Triplet Theory and History-Dependent Stochastic Processes: Fixed-State Indivisibility, Markov Order, and the Quantum Boundary*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18254862>